

# Geometry



## Area and perimeter

- 1 A rectangle has the dimensions shown below. Find its area and perimeter.

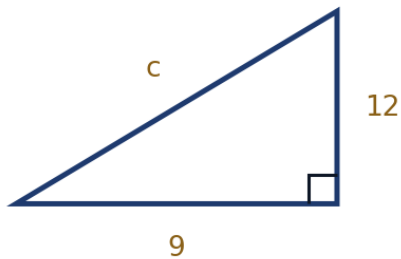


- 2 Find the area of a trapezoid with parallel sides 8 and 14, and height 6.
- 3 A triangle has base 15 and height 10. Find its area.

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## The Pythagorean theorem

- 4 A right triangle has the two legs shown below. Find the length of the hypotenuse  $c$ .



- 5 A 25-foot ladder leans against a wall, with its base 7 feet from the wall. How high up the wall does the ladder reach?

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## Volume and surface area

- 6 Find the volume of a cylinder with radius 4 and height 9, in terms of  $\pi$ .
- 7 Find the surface area of a rectangular prism with dimensions 5 by 4 by 3.

## Similar triangles and proportional reasoning

- 8 Triangle  $ABC$  is similar to triangle  $DEF$ , with  $AB = 8$ ,  $DE = 12$ , and  $BC = 10$ . Find  $EF$ .
- 9 A flagpole casts a shadow 18 feet long, while a 5-foot student standing nearby casts a shadow 3 feet long. Find the height of the flagpole.
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## Angle relationships

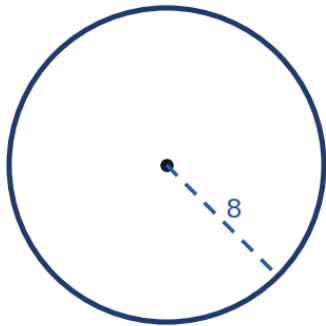
- 10 Two parallel lines are cut by a transversal. If one angle measures  $72^\circ$ , find its corresponding angle and its co-interior (same-side interior) angle.
- 11 The three angles of a triangle are in the ratio  $2 : 3 : 4$ . Find each angle.
- 12 Two angles are complementary, and one is  $34^\circ$  more than the other. Find both angles.
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## Coordinate geometry: distance and midpoint

- 13 Find the distance between the points  $(3, -2)$  and  $(9, 6)$ .
- 14 Find the midpoint of the segment connecting  $(-5, 3)$  and  $(7, -9)$ .
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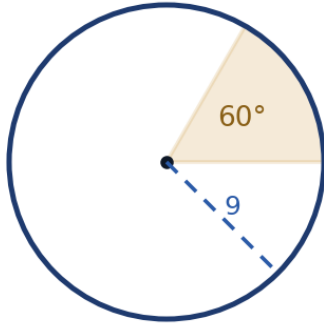
## Circles: area, circumference, and equation

- 15 A circle has the radius shown below. Find its area and circumference, in terms of  $\pi$ .



- 16 Write the equation of a circle with center  $(3, -4)$  and radius 6.

- 17 A circle has the radius and shaded sector shown below. Find the area of the shaded  $60^\circ$  sector, in terms of  $\pi$ .



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### Composite and shaded regions

- 18 A square with side length 12 has a circle of radius 5 removed from its center. Find the remaining area, in terms of  $\pi$ .
- 19 A rectangular garden 20 m by 15 m has a circular fountain of radius 3 m built inside it. Using  $\pi \approx 3.14$ , find the area of the garden not covered by the fountain, to the nearest whole number.

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### A well-designed word problem: designing a park

- 20 This question has three parts. A city is designing a park shaped like a rectangle with a semicircular section attached to one end. The rectangular section is 30 meters long and 20 meters wide, and the semicircle is attached to one of the 20-meter-wide ends. (a) Find the radius of the semicircle. (b) Find the total area enclosed by the park (rectangle plus semicircle), in terms of  $\pi$ . (c) The city fences the perimeter of the park: the three straight sides of the rectangle plus the curved semicircular edge, but not the shared 20-meter edge where the semicircle attaches. Find the total fence length needed, in terms of  $\pi$ .